In [ ]:
```
EXPERIMENT 8 – DATA PRE-PROCESSING & FEATURE ENGINEERING
PIN NO:-241UDS0109
AIM:-
To implement data pre-processing and feature engineering techniques on a dataset, including removing
unnecessary features, feature scaling, and encoding categorical variables using Python libraries.
```

In [ ]:
```
## Step 1: Import the Basic Libraries
```

In [1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

In [ ]:
```
## Step 2: Load & Analyze the Dataset
Load the CSV file from the same folder as this notebook.
```

In [2]:
```
df_241UDS0109 = pd.read_csv("Experiment_8_241UDS0109_Training_Performance.csv")
df_241UDS0109.head()
```

Out[2]:

Columns 1–8 of 22

	PIN	Branch	Marks[10th]	Marks[12th]	Gender	Board[10th]	Board[12th]
0	241UDS0109	ECE	70.08	57.05	Female	ICSE	ICSE
1	241UDS0109	ECE	99.00	89.22	Male	ICSE	ICSE
2	241UDS0109	CSE	88.13	89.76	Male	CBSE	CBSE
3	241UDS0109	MECH	72.87	61.60	Male	CBSE	CBSE
4	241UDS0109	ECE	79.46	81.14	Female	CBSE	CBSE

Columns 9–15 of 22 · first column repeated

	Category	GPA 1	Rank	...	GPA 3	GPA 4	GPA 5
0	SC	5.65	4706.0	...	4.28	4.15	4.00
1	OBC	7.92	8510.0	...	8.16	8.25	8.11
2	OBC	9.12	9907.0	...	8.37	8.63	8.30
3	OBC	6.43	41124.0	...	5.45	4.89	4.65
4	SC	8.03	27103.0	...	8.67	8.59	8.97

Columns 16–22 of 22 · first column repeated

	GPA 6	Olympiads Qualified	Technical Projects	Tech Quiz	Engg. Coaching	NTSE Scholarships	Miscellany Tech Events
0	4.28	False	2.0	False	False	NaN	3.0
1	9.34	True	1.0	True	True	True	4.0
2	8.65	True	0.0	False	True	False	1.0
3	4.92	False	4.0	True	False	False	5.0
4	8.78	False	6.0	False	False	True	3.0

5 rows × 25 columns

In [ ]:
```
### Dataset Information
```

In [3]:
```
df_241UDS0109.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 92 entries, 0 to 91
Data columns (total 25 columns):
 #   Column                  Non-Null Count  Dtype  
---  ------                  --------------  -----  
 0   PIN                     90 non-null     object 
 1   Branch                  88 non-null     object 
 2   Marks[10th]             88 non-null     float64
 3   Marks[12th]             88 non-null     float64
 4   Gender                  88 non-null     object 
 5   Board[10th]             88 non-null     object 
 6   Board[12th]             88 non-null     object 
 7   Category                88 non-null     object 
 8   GPA 1                   88 non-null     float64
 9   Rank                    90 non-null     float64
 10  Normalized Rank         90 non-null     float64
 11  CGPA                    88 non-null     float64
 12  Current Back            88 non-null     float64
 13  Ever Back               88 non-null     float64
 14  GPA 2                   88 non-null     float64
 15  GPA 3                   88 non-null     float64
 16  GPA 4                   88 non-null     float64
 17  GPA 5                   88 non-null     float64
 18  GPA 6                   88 non-null     float64
 19  Olympiads Qualified     88 non-null     object 
 20  Technical Projects      88 non-null     float64
 21  Tech Quiz               88 non-null     object 
 22  Engg. Coaching          88 non-null     object 
 23  NTSE Scholarships       88 non-null     object 
 24  Miscellany Tech Events  88 non-null     float64
dtypes: float64(15), object(10)
memory usage: 18.1+ KB
```

In [ ]:
```
### Check the Shape
```

In [4]:
```
df_241UDS0109.shape
```

Out[4]:
```
(92, 25)
```

In [ ]:
```
### Check for Null Values
```

In [5]:
```
df_241UDS0109.isnull().sum()
```

Out[5]:
```
PIN                     2
Branch                  4
Marks[10th]             4
Marks[12th]             4
Gender                  4
Board[10th]             4
Board[12th]             4
Category                4
GPA 1                   4
Rank                    2
Normalized Rank         2
CGPA                    4
Current Back            4
Ever Back               4
GPA 2                   4
GPA 3                   4
GPA 4                   4
GPA 5                   4
GPA 6                   4
Olympiads Qualified     4
Technical Projects      4
Tech Quiz               4
Engg. Coaching          4
NTSE Scholarships       4
Miscellany Tech Events  4
dtype: int64
```

In [8]:
```
df_241UDS0109 = df_241UDS0109.dropna(how="all").copy()

numeric_cols_241UDS0109 = df_241UDS0109.select_dtypes(include=np.number).columns
categorical_cols_241UDS0109 = df_241UDS0109.select_dtypes(exclude=np.number).columns

for col in numeric_cols_241UDS0109:
    df_241UDS0109[col] = df_241UDS0109[col].fillna(df_241UDS0109[col].median())

for col in categorical_cols_241UDS0109:
    mode_values = df_241UDS0109[col].mode(dropna=True)
    if not mode_values.empty:
        df_241UDS0109[col] = df_241UDS0109[col].fillna(mode_values.iloc[0]).infer_objects(copy=False)

df_241UDS0109.shape

# Restore boolean feature columns that may load as text because of missing values
boolean_features_241UDS0109 = ["Olympiads Qualified", "Tech Quiz", "Engg. Coaching", "NTSE Scholarships"]
for col in boolean_features_241UDS0109:
    df_241UDS0109[col] = df_241UDS0109[col].astype(str).str.lower().map({"true": True, "false": False})
```

In [ ]:
```
## Step 3: Remove Unnecessary Features
Remove the Rank column, as shown in the source experiment, then inspect the updated shape and columns.
```

In [9]:
```
df_241UDS0109 = df_241UDS0109.drop("Rank", axis=1)
df_241UDS0109.shape
```

Out[9]:
```
(90, 24)
```

In [10]:
```
df_241UDS0109.columns
```

Out[10]:
```
Index(['PIN', 'Branch', 'Marks[10th]', 'Marks[12th]', 'Gender', 'Board[10th]',
       'Board[12th]', 'Category', 'GPA 1', 'Normalized Rank', 'CGPA',
       'Current Back', 'Ever Back', 'GPA 2', 'GPA 3', 'GPA 4', 'GPA 5',
       'GPA 6', 'Olympiads Qualified', 'Technical Projects', 'Tech Quiz',
       'Engg. Coaching', 'NTSE Scholarships', 'Miscellany Tech Events'],
      dtype='object')
```

In [ ]:
```
## Step 4: Scale the Numerical Features
### Apply StandardScaler to Marks[10th] and Marks[12th]
```

In [11]:
```
from sklearn.preprocessing import StandardScaler
z_scaler_241UDS0109 = StandardScaler()
```

In [12]:
```
df_241UDS0109[["Marks[10th]", "Marks[12th]"]] = z_scaler_241UDS0109.fit_transform(
    df_241UDS0109[["Marks[10th]", "Marks[12th]"]]
)
df_241UDS0109.head(3)
```

Out[12]:

Columns 1–8 of 22

	PIN	Branch	Marks[10th]	Marks[12th]	Gender	Board[10th]	Board[12th]
0	241UDS0109	ECE	-0.686710	-1.393495	Female	ICSE	ICSE
1	241UDS0109	ECE	2.291344	1.347132	Male	ICSE	ICSE
2	241UDS0109	CSE	1.171999	1.393135	Male	CBSE	CBSE

Columns 9–15 of 22 · first column repeated

	Category	GPA 1	Normalized Rank	...	GPA 3	GPA 4	GPA 5
0	SC	5.65	9.412	...	4.28	4.15	4.00
1	OBC	7.92	17.020	...	8.16	8.25	8.11
2	OBC	9.12	19.814	...	8.37	8.63	8.30

Columns 16–22 of 22 · first column repeated

	GPA 6	Olympiads Qualified	Technical Projects	Tech Quiz	Engg. Coaching	NTSE Scholarships	Miscellany Tech Events
0	4.28	False	2.0	False	False	True	3.0
1	9.34	True	1.0	True	True	True	4.0
2	8.65	True	0.0	False	True	False	1.0

3 rows × 24 columns

In [ ]:
```
### Apply Normalization using MinMaxScaler to CGPA and GPA 1
```

In [13]:
```
from sklearn.preprocessing import MinMaxScaler
mm_scaler_241UDS0109 = MinMaxScaler()
```

In [14]:
```
df_241UDS0109[["CGPA", "GPA 1"]] = mm_scaler_241UDS0109.fit_transform(
    df_241UDS0109[["CGPA", "GPA 1"]]
)
df_241UDS0109.head(3)
```

Out[14]: Columns 1–8 of 22

	PIN	Branch	Marks[10th]	Marks[12th]	Gender	Board[10th]	Board[12th]
0	241UDS0109	ECE	-0.686710	-1.393495	Female	ICSE	ICSE
1	241UDS0109	ECE	2.291344	1.347132	Male	ICSE	ICSE
2	241UDS0109	CSE	1.171999	1.393135	Male	CBSE	CBSE

Columns 9–15 of 22 · first column repeated

	Category	GPA 1	Normalized Rank	...	GPA 3	GPA 4	GPA 5
0	SC	0.247423	9.412	...	4.28	4.15	4.00
1	OBC	0.715464	17.020	...	8.16	8.25	8.11
2	OBC	0.962887	19.814	...	8.37	8.63	8.30

Columns 16–22 of 22 · first column repeated

	GPA 6	Olympiads Qualified	Technical Projects	Tech Quiz	Engg. Coaching	NTSE Scholarships	Miscellany Tech Events
0	4.28	False	2.0	False	False	True	3.0
1	9.34	True	1.0	True	True	True	4.0
2	8.65	True	0.0	False	True	False	1.0

3 rows × 24 columns

In [ ]:
```
## Step 5: Encode the Categorical Features
### Apply Integer Encoding using LabelEncoder to Gender, Category, and Branch
```

In [15]:
```
from sklearn.preprocessing import LabelEncoder
encoder_241UDS0109 = LabelEncoder()
```

In [16]:
```
for col in ["Gender", "Category", "Branch"]:
    df_241UDS0109[col] = encoder_241UDS0109.fit_transform(df_241UDS0109[col].astype(str))

df_241UDS0109.head(3)
```

Out[16]: Columns 1–8 of 22

	PIN	Branch	Marks[10th]	Marks[12th]	Gender	Board[10th]	Board[12th]
0	241UDS0109	2	-0.686710	-1.393495	0	ICSE	ICSE
1	241UDS0109	2	2.291344	1.347132	1	ICSE	ICSE
2	241UDS0109	1	1.171999	1.393135	1	CBSE	CBSE

Columns 9–15 of 22 · first column repeated

	Category	GPA 1	Normalized Rank	...	GPA 3	GPA 4	GPA 5
0	2	0.247423	9.412	...	4.28	4.15	4.00
1	1	0.715464	17.020	...	8.16	8.25	8.11
2	1	0.962887	19.814	...	8.37	8.63	8.30

Columns 16–22 of 22 · first column repeated

	GPA 6	Olympiads Qualified	Technical Projects	Tech Quiz	Engg. Coaching	NTSE Scholarships	Miscellany Tech Events
0	4.28	False	2.0	False	False	True	3.0
1	9.34	True	1.0	True	True	True	4.0
2	8.65	True	0.0	False	True	False	1.0

3 rows × 24 columns

In [ ]:
```
### Apply One-Hot Encoding using pd.get_dummies() to Board[10th] and Board[12th]
```

In [17]:
```
df_241UDS0109 = pd.get_dummies(
    df_241UDS0109,
    columns=["Board[10th]", "Board[12th]"],
    drop_first=True
)
df_241UDS0109.head(3)
```

Out[17]:

Columns 1–8 of 22

	PIN	Branch	Marks[10th]	Marks[12th]	Gender	Category	GPA 1
0	241UDS0109	2	-0.686710	-1.393495	0	2	0.247423
1	241UDS0109	2	2.291344	1.347132	1	1	0.715464
2	241UDS0109	1	1.171999	1.393135	1	1	0.962887

Columns 9–15 of 22 · first column repeated

	Normalized Rank	CGPA	Current Back	...	Olympiads Qualified	Technical Projects	Tech Quiz
0	9.412	0.005736	1.0	...	False	2.0	False
1	17.020	0.736138	2.0	...	True	1.0	True
2	19.814	0.782027	1.0	...	True	0.0	False

Columns 16–22 of 22 · first column repeated

	Engg. Coaching	NTSE Scholarships	Miscellany Tech Events	Board[10th]_ICSE	Board[10th]_State Board	Board[12th]_ICSE	Board[12th]_State Board
0	False	True	3.0	True	False	True	False
1	True	True	4.0	True	False	True	False
2	True	False	1.0	False	False	False	False

3 rows × 26 columns

In [ ]:
```
### Check DataFrame Information after Encoding
```

In [18]:
```
df_241UDS0109.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 90 entries, 0 to 89
Data columns (total 26 columns):
 #   Column                   Non-Null Count  Dtype
---  ------                   --------------  -----
 0   PIN                      90 non-null     object
 1   Branch                   90 non-null     int64
 2   Marks[10th]              90 non-null     float64
 3   Marks[12th]              90 non-null     float64
 4   Gender                   90 non-null     int64
 5   Category                 90 non-null     int64
 6   GPA 1                    90 non-null     float64
 7   Normalized Rank          90 non-null     float64
 8   CGPA                     90 non-null     float64
 9   Current Back             90 non-null     float64
 10  Ever Back                90 non-null     float64
 11  GPA 2                    90 non-null     float64
 12  GPA 3                    90 non-null     float64
 13  GPA 4                    90 non-null     float64
 14  GPA 5                    90 non-null     float64
 15  GPA 6                    90 non-null     float64
 16  Olympiads Qualified      90 non-null     bool
 17  Technical Projects       90 non-null     float64
 18  Tech Quiz                90 non-null     bool
 19  Engg. Coaching           90 non-null     bool
 20  NTSE Scholarships        90 non-null     bool
 21  Miscellany Tech Events   90 non-null     float64
 22  Board[10th]_ICSE         90 non-null     bool
 23  Board[10th]_State Board  90 non-null     bool
 24  Board[12th]_ICSE         90 non-null     bool
 25  Board[12th]_State Board  90 non-null     bool
dtypes: bool(8), float64(14), int64(3), object(1)
memory usage: 14.1+ KB
```

In [ ]:
```
### Convert Boolean Datatypes into Integers
```

In [19]:
```
bool_columns_241UDS0109 = df_241UDS0109.select_dtypes(include="bool").columns

for col in bool_columns_241UDS0109:
    df_241UDS0109[col] = df_241UDS0109[col].astype(int)

df_241UDS0109.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 90 entries, 0 to 89
Data columns (total 26 columns):
 #   Column                   Non-Null Count  Dtype
---  ------                   --------------  -----
 0   PIN                      90 non-null     object
 1   Branch                   90 non-null     int64
 2   Marks[10th]              90 non-null     float64
 3   Marks[12th]              90 non-null     float64
 4   Gender                   90 non-null     int64
 5   Category                 90 non-null     int64
 6   GPA 1                    90 non-null     float64
 7   Normalized Rank          90 non-null     float64
 8   CGPA                     90 non-null     float64
 9   Current Back             90 non-null     float64
 10  Ever Back                90 non-null     float64
 11  GPA 2                    90 non-null     float64
 12  GPA 3                    90 non-null     float64
 13  GPA 4                    90 non-null     float64
 14  GPA 5                    90 non-null     float64
 15  GPA 6                    90 non-null     float64
 16  Olympiads Qualified      90 non-null     int64
 17  Technical Projects       90 non-null     float64
 18  Tech Quiz                90 non-null     int64
 19  Engg. Coaching           90 non-null     int64
 20  NTSE Scholarships        90 non-null     int64
 21  Miscellany Tech Events   90 non-null     float64
 22  Board[10th]_ICSE         90 non-null     int64
 23  Board[10th]_State Board  90 non-null     int64
 24  Board[12th]_ICSE         90 non-null     int64
 25  Board[12th]_State Board  90 non-null     int64
dtypes: float64(14), int64(11), object(1)
memory usage: 19.0+ KB
```

In [ ]:
```
### Check Final Shape
```

In [20]:
```
df_241UDS0109.shape
```

Out[20]:
```
(90, 26)
```

In [ ]:
```
### Save the Processed Dataset as a CSV File
```

In [21]:
```
df_241UDS0109.to_csv(
    "processed_training_dataset_241UDS0109.csv",
    index=False
)
print("Processed dataset saved successfully.")
print("Final shape:", df_241UDS0109.shape)
```

```
Processed dataset saved successfully.
Final shape: (90, 26)
```

In [ ]:
```
RESULT:-
The training-performance dataset was pre-processed by removing fully empty rows and an unnecessary Rank
feature, filling remaining missing values, scaling numerical features using StandardScaler and
MinMaxScaler, and encoding categorical features using LabelEncoder and one-hot encoding. Boolean columns
were converted into integers, and the processed dataset was saved as a CSV file.
```